

TSL-51: Thai Sign Language Recognition System

TSL Research Team

April 2026

Chapter 1

1.1

(Sign Language) (Thai Sign Language: TSL)
TSL-51 51

1.2

1. TSL-51
2. TUI
- 3.
- 4.

1.3

- : 51
- : TSL-51 Dataset
- : GRU MLP
- Features: 162

Chapter 2

2.1 Deep Learning

Deep Learning Machine Learning Neural Network

2.2 GRU

Gated Recurrent Unit (GRU) variant RNN

2.3 MediaPipe

MediaPipe Framework Google Landmark

Chapter 3

3.1

TSL-51 :

-
-
- Inference
- TUI

3.2 GRU

```
1 class GRUModel(nn.Module):
2     def __init__(self, input_dim=162, hidden_dim=256,
3                   num_classes=51, num_layers=3, dropout=0.3):
4         super().__init__()
5         self.gru = nn.GRU(
6             input_size=input_dim,
7             hidden_size=hidden_dim,
8             num_layers=num_layers,
9             batch_first=True,
10            bidirectional=True,
11        )
12        self.fc = nn.Linear(hidden_dim * 2, num_classes)
```

Chapter 4

4.1

	Accuracy	F1-Score
GRU (3 layers)	92.5%	0.91
MLP (3 layers)	87.3%	0.86

4.2

GRU MLP

Chapter 5

5.1

TSL-51 :

- 92.5% GRU
- TUI
- Real-time Camera Translation

5.2

- 1.
2. Sentence-level Translation
3. Real-time Performance

Bibliography

Google. (2024). MediaPipe. <https://mediapipe.dev>
Namonpas. (2024). Thai Sign Language TSL-51 Dataset. HuggingFace.
Cho, K., et al. (2014). Learning Phrase Representations using RNN Encoder-Decoder. EMNLP 2014.
Paszke, A., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. NeurIPS 2019.